

Lab: Functions in Python

Problem Statement 1: Student Grade Calculator

Write a Python program that defines a function `calculate_grade(marks)`.

The function should:

- Accept marks (0–100) as a parameter.
- Return the grade according to the following criteria:
 - 90 and above → A+
 - 75–89 → A
 - 60–74 → B
 - 40–59 → C
 - Below 40 → Fail

The main program should:

- Accept marks of 5 students.
 - Call the function for each student.
 - Display the marks and corresponding grade.
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Problem Statement 2: List Analysis using Functions

Write a Python program that defines the following functions:

- `find_max(numbers)`
- `find_min(numbers)`
- `find_average(numbers)`

The program should:

- Accept a list of 10 integers from the user.
 - Call all three functions.
 - Display the maximum value, minimum value, and average of the list.
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Problem Statement 3: Password Strength Checker

Write a function `check_password(password)` that checks whether a password is strong.

A password is considered **Strong** if:

- It contains at least 8 characters.
- It contains at least one uppercase letter.

- It contains at least one lowercase letter.
- It contains at least one digit.

The function should return:

- "Strong Password" or
- "Weak Password"

The main program should accept a password from the user and display the result.

Problem Statement 4: Dictionary Search System

Write a Python program that defines a function `search_student(student_dict, roll_no)`.

The function should:

- Accept a dictionary where:
 - Key = Roll Number
 - Value = Student Name
- Search for the given roll number.
- Return the student name if found; otherwise return "Student Not Found".

The main program should:

- Create a dictionary of at least 5 students.
 - Accept a roll number from the user.
 - Display the search result.
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Problem Statement 5: Vowel Counter using Function

Write a Python program that defines a function `count_vowels(text)`.

The function should:

- Accept a string.
- Count the total number of vowels (a, e, i, o, u) irrespective of case.
- Return the total vowel count.

The main program should:

- Accept a sentence from the user.
- Call the function.
- Display the total number of vowels.

Sample Output

Enter a sentence:

Python Programming is Fun

Total Vowels: 6